



Ordinary Differential Equations

First-Order Differential Equations (Solutions by Substitutions)

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ENGR 213**

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Today's Lecture

- Substitution Methods
 - ✓ Homogeneous Equations
 - ✓ Bernoulli's Equation
 - ✓ Reduction to Separation of Variables

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 2.5**

- Analytical methods for first-order ODEs:

- Separable Equations $\longrightarrow \frac{dy}{dx} = g(x)h(y)$

- Linear Equations $\longrightarrow a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$

- Exact Equations $\longrightarrow M(x, y)dx + N(x, y)dy = 0$

e.g.: $\frac{dy}{dx} = f(x, y) = 1 + \frac{y}{x} + \frac{y^2}{x^2}$

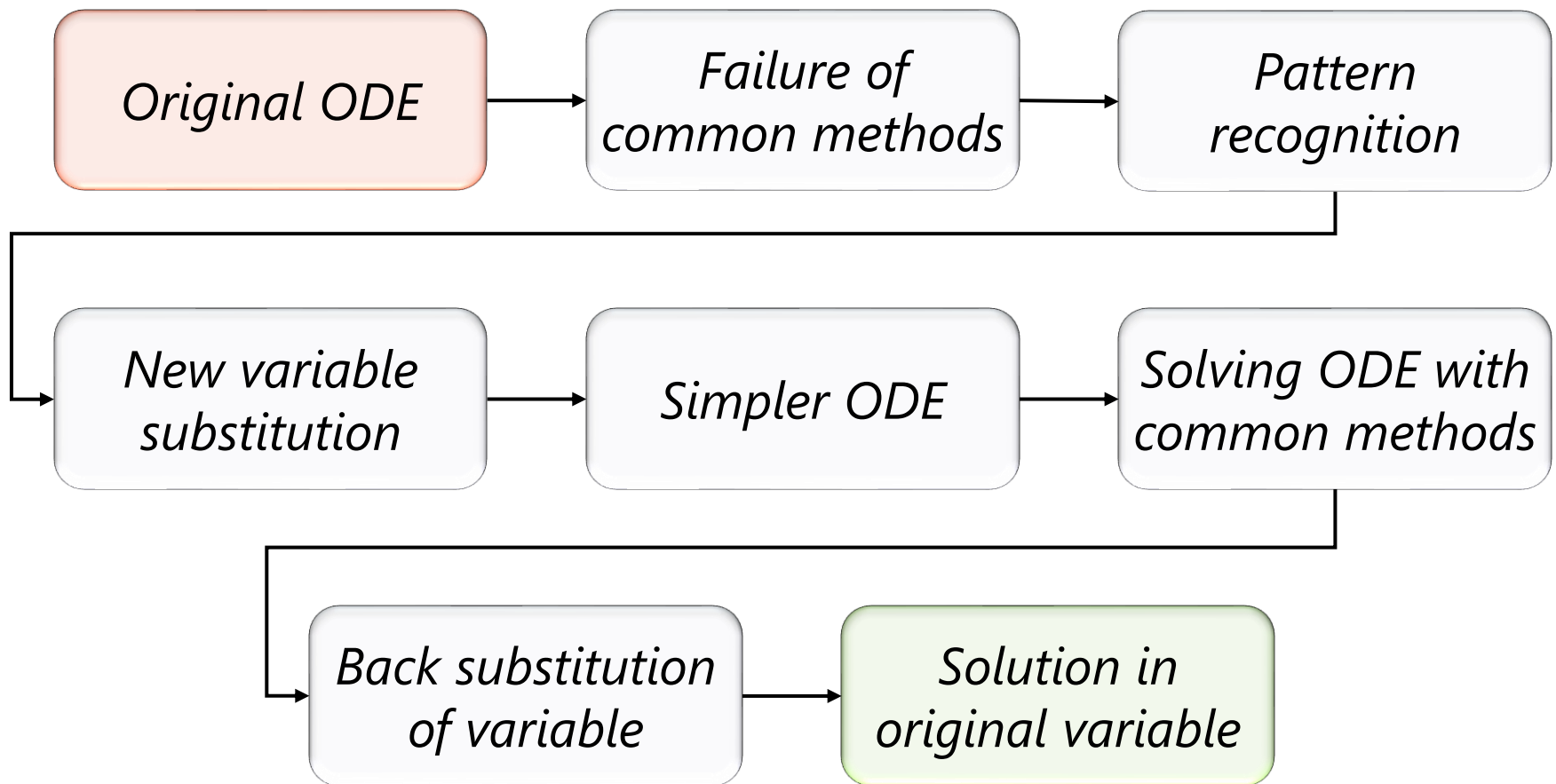
✓ **Can we solve this using one of the methods from our previous lectures?**

Solutions by Substitutions

- Since this ODE doesn't appear to fit a familiar form, we look for a change of variable that might reveal a simpler, solvable structure.

e.g.: $\frac{dy}{dx} = 1 + \frac{y}{x} + \frac{y^2}{x^2}$





The goal of substitution is not to create a new solution method, but to convert the ODE into a familiar solvable form, such as a separable or linear equation.

EXAMPLE 1 (pattern recognition)

$$\frac{dy}{dx} = (x + y)^2$$

$$\frac{dy}{dx} = 2 + e^{2x-y+1}$$

$$\frac{dy}{dx} + 2y = xy^3$$

$$\frac{dy}{dx} = y(x + \ln y)$$

□ Homogeneous Equations

- If function f has the property $f(tx, ty) = t^\alpha f(x, y)$ for some real number α , then f is a homogeneous function of degree α .

e.g.: $f(x, y) = x^3 + y^3$

- A first-order ODE written in differential form

$$M(x, y)dx + N(x, y)dy = 0$$

is a homogeneous equation if M and N are homogeneous functions of the same degree.

- Either of substitutions $u = \frac{y}{x}$ or $v = \frac{x}{y}$ will reduce the homogeneous equation to a separable 1st order ODE.

EXAMPLE 2

Solve $(x^2 + y^2)dx + (x^2 - xy)dy = 0$.

□ Bernoulli's Equation

- The differential equation $\frac{dy}{dx} + P(x)y = f(x)y^n$ where n is any real number, is called Bernoulli's equation.

- Some special cases of Bernoulli:

$$n = 0: \quad y' + P(x)y = f(x) \quad \text{linear 1}^{\text{st}}\text{-order ODE}$$

$$n = 1: \quad y' + P(x)y = f(x)y \quad \text{Separable 1}^{\text{st}}\text{-order ODE}$$

$$n \neq 0 \text{ and } n \neq 1: \quad y' + P(x)y = f(x)y^n \quad \text{Nonlinear ODE}$$



This can be reduced to a linear
ODE by substituting $u = y^{1-n}$

$$u' + (1 - n)P(x)u = (1 - n)f(x) \quad \text{linear 1}^{\text{st}}\text{-order ODE in } u$$

EXAMPLE 3

Solve $x \frac{dy}{dx} + y = x^2 y^2$.

□ Reduction to Separation of Variables

- A differential equation of the form $\frac{dy}{dx} = f(Ax + By + C)$

can always be reduced to an equation with separable variables by means of the substitution $u = Ax + By + C$, $B \neq 0$.

EXAMPLE 4

Solve $\frac{dy}{dx} = (-2x + y)^2 - 7$.